

AI Distributed System For Resource Optimizing Allocation in Cloud Infrastructure

A. Thameem Ansari*
B.E.CSE
Department of Computer Science
Engineering
Rajalakshmi Institute of Technology
Chennai, India
thameemansari.a.2021.cse@ritchennai.edu.in

K.S. Thirumurugan
B.E.CSE
Department of Computer Science
Engineering
Rajalakshmi Institute of Technology
Chennai, India
thirumurugan.ks.2021.cse@ritchennai.edu.in

AG. Noorul Julaiha
M.E.CSE Assistant Professor
Department of Computer Science
Engineering
Rajalakshmi Institute of Technology
Chennai, India
ag.nooruljulaiha@gmail.com

Abstract—Globally, the efficient management of cloud resources is critical as organizations increasingly rely on cloud computing to support diverse and dynamic applications. As cloud infrastructures expand, the challenge of resource allocation becomes more pronounced, particularly due to fluctuating workloads that necessitate responsive and adaptable solutions. Traditional static allocation methods often lead to inefficiencies, resulting in either resource wastage or inadequate provisioning, affecting performance and cost-efficiency. This paper proposes an AI-driven framework designed to optimize resource allocation in large-scale cloud environments. By integrating advanced machine learning techniques, including predictive modeling and reinforcement learning, the framework dynamically adjusts computational resources—such as CPU, memory, and bandwidth—based on real-time workload demands. The methodology involves a comprehensive data pipeline that ingests historical and current resource usage metrics, which are subsequently analyzed to forecast future needs. A reinforcement learning agent optimizes resource allocations dynamically by applying an enhanced gradient-based RL algorithm, allowing for a proactive response to workload fluctuations. Evaluation results indicate that our framework achieves a 27.3% reduction in operational costs compared to traditional methods, significantly improved SLA compliance, and reduced energy consumption by 31.6%. Additionally, the system demonstrates a 36.1% improvement in response time and an average CPU utilization of 73.1%, showcasing its effectiveness in adapting to varying cloud environments. This innovative approach not only fulfills the operational demands of modern cloud infrastructures but also sets the stage for future advancements in intelligent cloud resource management strategies, particularly in hybrid cloud-edge systems.

Index Terms—Cloud Computing, AI Optimization, Distributed Systems